

0.33

24. Answer: A

Taking moment about the top.

Total clockwise moment = Total anticlockwise moment

$$NBIL \times L \cos 30^\circ = 0.12 \times \frac{L}{2} \times \sin 30^\circ$$

$$50 \times B \times 0.40 \times 60 \times 10^{-3} \times \cos 30^\circ = 0.12 \times \frac{1}{2} \times \sin 30^\circ$$

$$B = 0.029 \text{ T}$$